FTP F1 (FUZE Textile Product)

Technical Data Sheet

Product Description

FTP F1 (FUZE Textile Product) is a textile treatment designed for integration during standard textile finishing operations. It can be applied using conventional pad-bath, exhaust, or spray processing equipment. The treatment integrates within the textile structure while maintaining the original fabric characteristics. FTP F1 is the highest-performance tier in the FUZE product range, targeting 1.0 mg/kg active loading on treated fabric for antimicrobial performance.

Product ID: FTP F1 **Document ID:** TDS-FTP-F1-002 **Revision:** 1.1 **Effective:** 2026-09-24 **Supersedes:** Rev 1.0

Product Properties

Property	Value
Appearance	Clear, colorless liquid
Carrier	18 megaohm DI water (99.997% w/w)
Active concentration	30 mg/L (0.003% w/w)
pH	~7 (neutral) as supplied — see Nylon Application Protocol for bath pH adjustment at point of use
Odor / Flavor	None / None
Reactivity	Non-reactive with textile finishing chemistries
Solubility	Fully miscible in water
Viscosity	Similar to water (~1.0 cP)
Density	~1.0 g/mL
Packaging	19 L HDPE container

Recommended Product Concentration in Textile Recipe

Pad Application — 80% Dry-to-Wet Pickup (typical pad process)

Tier	Target (mg/kg WOF)	Bath Conc. (mg/L)	Dosage (mL/L bath)	per 100 L Bath
F1	1.00	1.25	41.7	4.17 L
F2	0.75	0.94	31.2	3.12 L
F3	0.50	0.62	20.8	2.08 L
F4	0.25	0.31	10.4	1.04 L

Pad Application — 15% Wet-to-Wet Pickup (pre-wetted fabric)

Tier	Target (mg/kg WOF)	Bath Conc. (mg/L)	Dosage (mL/L bath)	per 100 L Bath
F1	1.00	6.67	222.2	22.22 L
F2	0.75	5.00	166.7	16.67 L
F3	0.50	3.33	111.1	11.11 L
F4	0.25	1.67	55.6	5.56 L

Calculation method: Bath concentration (mg/L) = Target loading (mg/kg) ÷ (pickup % ÷ 100). Dosage (mL per liter of bath) = Bath concentration (mg/L) ÷ stock concentration × 1000. Dosage per 100 L bath = Dosage (mL/L) × 100 ÷ 1000. Reference fabric 100 GSM; GSM cancels out of the bath-concentration formula. Adjust for fabric weight/construction and confirm by mill trial.

Fabric Compatibility

FTP F1 can be applied to polyester, nylon, polyester blends, cotton blends, synthetic performance fabrics, and down-proof fabrics. Compatibility with specialty finishing chemistries should be confirmed through standard mill trials. **For nylon and polyamide blends, follow the Nylon Application Protocol below** — bath pH adjustment materially improves exhaustion and rinse-fastness on polyamide. Polyester and cotton are applied at neutral bath pH.

Application Methods

Pad-Dry (Continuous Finishing)

Prepare bath with water carrier and add FTP F1 at the dosage indicated for the target loading and expected pickup. Pad fabric to achieve 10–90% wet pickup depending on fabric construction. Dry fabric using standard mill drying conditions. No curing step required.

Exhaust (Batch Processing)

Add FTP F1 to the dyebath or a dedicated finishing bath at the appropriate concentration. Run at standard dyeing temperature and time. FUZE integrates during the exhaust cycle. Compatible with existing dyebath chemistry — no process modification required.

Spray Application

Apply using spray heads spaced at 6-inch intervals across fabric width (top and bottom). Standard line speed: 15 m/min. Adjust spray concentration based on pump rate and target dosage. Suitable for continuous finishing lines.

Nylon Application Protocol (pH-Assisted Deposition)

Polyamide fibers carry amine end groups that become positively charged below approximately pH 5. The FTP F1 active carries a negative surface charge in water. Adjusting the application bath to pH 4.5–5.0 creates an electrostatic attraction between fiber and active that improves exhaustion onto nylon and rinse-fastness of the treatment. These are the same bath conditions used for standard acid-dyeing of nylon.

Step	Instruction
1	Prepare the bath with DI or softened water free of chloride. Do not use untreated tap water — chloride precipitates the active.
2	Add FTP F1 at the dosage for the target tier and expected pickup (tables above).
3	Adjust bath to pH 4.5–5.0 using acetic acid only (typically 0.3–0.5 mL/L of 10% acetic acid; verify with meter or strips). Never use hydrochloric acid or any chloride-containing acid. Do not reduce below pH 4.0. Do not add salts.
4	Apply within 60 minutes of acidification. Pad, exhaust, or spray and dry under standard mill conditions. No curing step required.
5	Fabric must be scoured and free of softeners and spin-finish lubricants before treatment.
6	Verify active loading by ICP against the tier target.

Important: Adjust pH in the application bath only, at point of use. Do not pre-acidify FTP F1 stock and do not store the product at reduced pH. Bath acidification is not required and is not recommended for polyester or cotton.

Storage and Handling

Storage: Store in original HDPE container, tightly closed. Temperature: 5°C–35°C. Avoid freezing.
Shelf life: 12 months when stored unopened.
Handling: Use standard industrial hygiene practices. No special PPE required under normal conditions.

Regulatory Information

FTP F1 contains elemental silver (CAS 7440-22-4) at 0.003% w/w. Not classified as hazardous under GHS at supplied concentration. Consult the Safety Data Sheet (SDS) for full regulatory and safety information.

FTP F1 is manufactured, delivered, applied, and stored exclusively as a dilute aqueous dispersion. The active is never supplied or made available in dry or free particulate form, and is not present on the finished textile as free nano-scale material available for interaction or release. On this basis the product is positioned as a non-nano textile treatment for regulatory purposes.

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